MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, AUDIO INTERFACE

This specification is approved for use by the US Army Armament , Munitions and Chemical Canmand and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manuf act ure and acceptance of the Audio Interface Circuit Card Assembly: 9355830, used in the Mortar Ballistics Compter (MBC): M23 Throughout this specification this assembly is referenced as " interface" .

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2. 1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense index of Specification and Standards (DoDISS) spcified in the solici tat ion form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection
MIL-I 45607	-	Inspection Equipment Acquisition, Maintenance and Disposition of

STANDARIE

MIL ITA RY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-454	-	Electronic Equipment, General Requirements for
MIL-STD-1521	-	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STD-45662	-	Calibration System Requirements

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal.

DRAWINGS

9355830	Circuit Card Assembly, Audio Interfac
11785850	Computer Set, Ballistics, Mortar: M23

PACKAGING DATA SHEETS

9355830	Packaging of Circuit Card Assembly, Audio Interfacd: 9355830

[Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The Audio Interface Circuit Card Assembly shall provide impedance matching and level shifting circuitry to interface with standard US Army communications systems (FM radios, AM radios, and wireline) . A block diagram of the major hardware functions is shown in Figure 1 and the connector contacts are listed in Table I.

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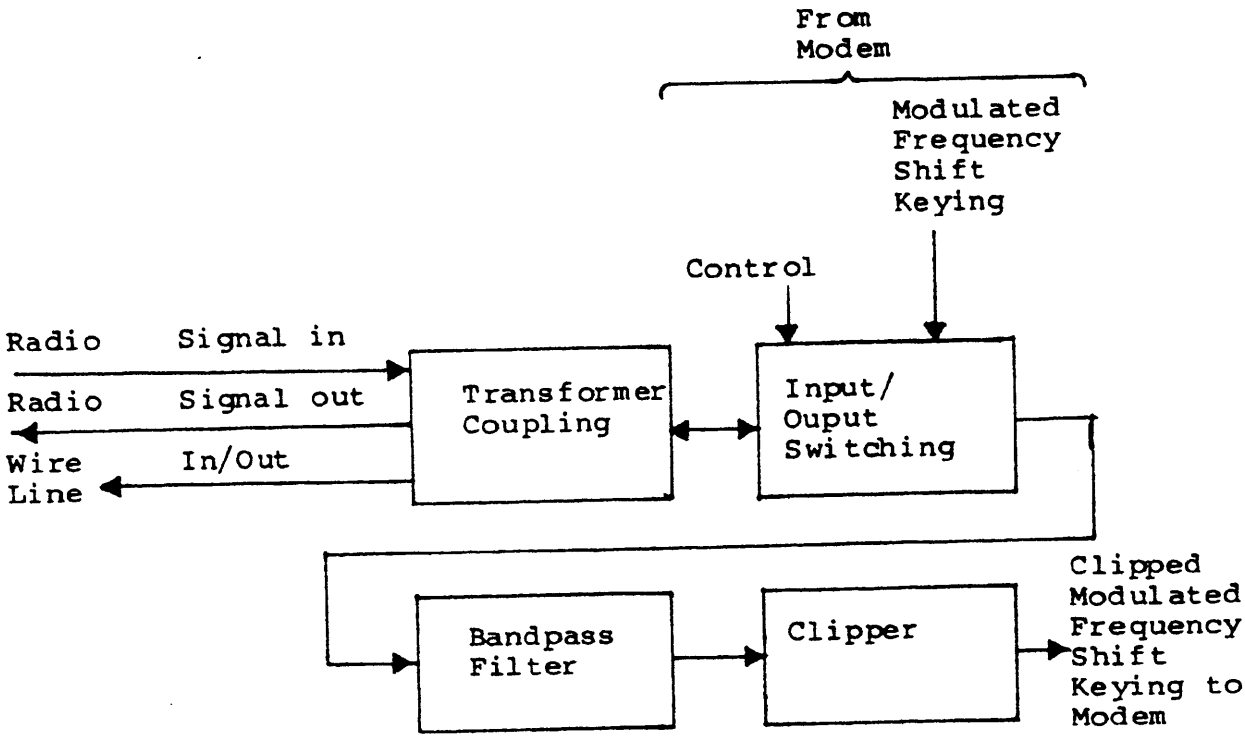


FIGURE 1. Audio interface block diagram.

TABLE I. Audio interface circuit card assembly connector contacts.

Connector	Pin	Type	Name
P1	1	O	CFSKO
P1	2	I	WXMT
P1	3		WIREN
P1	4	I	MFSK
P1	5	I	+5V
P1	6	I	-5V
P1	7		GND
P1	8		Spare
P1	9		XMTKYR
P1	10		XMTKY
J1	A1		XMTKY
J1	B1		XMTKYR
J1	A2		Spare
J1	B2		GND
J1	A3	O	AUDIO TO FM RADIO
J1	B3		WIREN
J1	A4		RADIO COMMON

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TABLE I. Audio interface circuit card assembly connector contacts.-Continued

Connector	Pin	TYPE	Name
J1	B4	0	AUDIO TO AM RADIO
J1	AS	I	AUDIO FROM AM RADIO
J1	B5	I	AUDIO FROM FM RADIO
J1	A6	I/o	WIRELINE
J1	B6	I/o	WIRELINE

3.2 Requirements.

3.2.1 Performance. The performance of the interface shall be in accordance with the following requirements.

3.2.1. Audio to FM radio. Connector J1-A3 shall be a +5 plus or minus 2 millivolt RMS sine wave when terminated into a 150 ohm plus or minus 1 percent resistor and referenced to J1-A4.

3.2.1.2 Audio to AM radio. Connector J1-B4 shall be a +19.5,5 plus or minus 4 millivolt rms sine wave when terminated into a 50 ohm plus or minus 1 percent resistor and referenced to J1-A4.

3.2.1.3 Wireline output. Connector J1-B6 shall be a +775 millivolt plus or minus 200 millivolt sine wave when terminated into a 600 ohm plus or minus 1 percent resistor and referenced to J1-A6.

3.2.1.4 Clipped frequency shift keying. Connector P1-l shall be a square wave with positive level greater than +3. 5 volts and a negative level less than minus 3.5 volts when referenced to Pi-7.

3.2.1.5 Radio input. With an +8 millivolt to 2 volt rms sine wave applied to J1-B5 with respect to J1-A4, Pi-l shall be as specified in 3.2.1.4.

3.2.1.6 Wireline input. With a +7.75 millivolt to .775 volt rms sine wave applied to J1-B6 with respect to J1-A6, P1-1 shall be as specified 3.2.1.4.

3.2. 1.7 Common connector contact continuity. The resistance between the following respective connector contacts shall be less than 7.5 ohms.

J1-A1 to P1-10
J1-B1 to P1-9
J1-B2 to P1-7
J1-B3 to P1-3
J1-B5 to J1-A5

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3.2.2 Physical characteristics. The physical characteristics shall as prescribed by the drawings, specifications? and QAP's.

3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques employed in the fabrication of the interface shall not degrade the inherent reliability. The inherent reliability shall not be less than 1,323,627 hours Mean Time Between Failures (MTBF). (See 4 and 6]

3.3 Environmental. The interface shall meet the requirements specified herein unless otherwise specified-

3.3.1 Room ambient temperature. The requirements of this specification shall be met with the interface thermally stabilized at room ambient temperature of +16°C to +32°C (+60°F to +90^{0F}) , unless otherwise specified in the requirements paragraph.

3.3.2 Environmental stress screen (ESS) temperature. The interface shall meet the requirements of 3.2.1.1 and 3.2.1.7 inclusive) after having been exposed to a minimum of ten temperature cycles at temperatures of -57°C and +71°C (-71°F and +160^{0}F) .

3.3.3 Environmental stress screen {ESS) vibration. The interface shall meet the requirements of 3.2.1.1 and 3.2.1.7 inclusive? after having been subjected to a random or pseudorandom vibration which includes frequency content from 100 Hz to 1000 Hz and up to a 39 RMS in two axis for a period of 10 minutes plus or minus 15 seconds.

3.4 Fabrication. The interface shall be manufactured in accordance with drawing 9355830.

3.5 First article. When specified in the contract, a sample shall be subjected to first article inspection (See 4 and 6).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall be met. All components shall be fitted properly and firmly in their respective positions. All electrical connections shall be mechanically secure and electrically sound. There shall be no evidence of corrosion, burrs, roughness, sharp edges other defects which could adversely affect its use or suitability for the purpose intended.

3.7 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsobility for inspection. Unless otherwise specified iin the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facili ties suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such i nspections are deemed necessary to assure suppli es and services conform to prescribed requi rements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13 926 AND MIL-STD-454 form a parrt of the Quality Assurance Provisions of this specification. defini tions of inspe ction terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) appproval. The requirement for first article approval and the respsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) interfaces plus three (3) each of all items covered by QAP. The three interfaces shall be tested as specified herein for all the requirements of this specification. The i terns covered by QAP shall be tested, as specified thereon, for all characters tics delineated on the QAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4. 2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4. 2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this i nspection, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractor’ s inspections .

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, Inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provision of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shallbe inspected. In accordance with the inspection provisions contained in Quality Assurance Provisions (QAP'sj listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4 Sampling.

4.4.1 Environmental/performance. The interface shall be inspected to the requirements in Table II.

TABLE II. Classification of defects.

Characteristics	Requirement	Test Procedure
100% Inspection		
1. Audio to FM radio	3.2.1.1	4.6.3.1
2. Audio to AM radio	3.2.1.2	4.6.3.2
3. Wireline output	3.2.1.3	4.6.3.3
4. Clipped frequency shift keying	3.2.1.4	4.6.3.4
5. Radio input	3.2.1.5	4.6.3.5
6. Wireline input	3.3.1.6	4.6.3.6
7. Common connector contact continuity	3.3.1.7	4.6.4
8. ESS temperature	3.3.2	4.6.1
9. ESS vibration	3.3.3	4.6.2
MAJOR: AQL 1.0% Defective		
101. Workmanship	3.6	4.6.5

4.4.2 Special sampling. One interface shall be selected at random as a special sample from each lot or each 100 produced which ever occurs first. The-sample shall meet the requirements and tests in Table III.

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TABLE III. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
201.	Fabrication	3.4	Applicable Drawing/ Visual
202.	Mate ri als	3.7	Applicable Drawing/ visual

NOTE: The tests in Table III shall be conducted at roan ambient temperature (as specified in 3.3.1).

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptace of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-145607.

4.5.1 Government furnished inspeection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-145607.

4.5.2 Contractor fUrnished inspeection equipment.

4.5.2.1 Government design All inspection equipment specified by drawing number In" specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List (TEL).

4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply inspction equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13 926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, instruction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance

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limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The interface shall be placed in the test chamber and the temperature of the chamber reduced 10°C (18°F) per minute plus or minus 1°C (1.8°F) from room ambient temperature (as specified in 3.3.1) to -57°c (-71°F). The interface shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the chamber shall be raised 10°c (18°F) per minute plus or minus l°C (1.8°F) to +71°F (+160°c)e The interface shall remain at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (18°F) per minute plus or minus 1° (1.8°F) to room

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ambient temperature (as specified in 3.3.1). The above procedure shall be conducted a minimum of ten times. Upon completion of the cycles the interface shall then be removed from the chamber and subjected to the tests in 4.6.3.1 and 4.6.4 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.7 inclusive.

4.6.2 ESS vibration. Conduct the vibration test with the interface positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. Unless otherwise specified, the test equipment shall be in accordance with the “Test Facilities” requirements of MIL-F-13926. The applied motion shall be in two separate planes for the time period specified. Upon completion of the vibration sweep the interface shall be removed from the vibration adapter and shall be subjected to the tests in 4.6.3.1 and 4.6.4 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.7 inclusive.

4.6.3 Performance. The performance test 4.6.3.1 and 4.6.3.6 inclusive, shall be Performed with +5.3 Plus or minus .1 volts DC applied to Pi-5 and minus 5.3 plus or minus .1 volts DC applied to P1-6 both with respect to P1-7.

4.6.3.1 Audio to FM radio. Apply +2.4 plus or minus 0.2 volt rms sine wave (1200 Hz) to P1-4 with respect to GND. The output response shall be as specified in 3.2.1.1.

4.6.3.2 Audio to AM radio. Apply a +2.4 plus or minus or minus 0.2 volt rms sine wave (1200 Hz) to P1-4 with respect to GND. The output response shall be as specified in 3.2.1.2.

4.6.3.3 Wireline output. Apply a +2.4 plus or minus 0.2 volt rms sine wave (1200 Hz) to PI-4 with respect to GND and connect P1-2 to P1-5. The output response shall be as specified in 3.2.1.3.

4.6.3.4 Clipped frequency shift keying. Apply a +2.4 plus or minus 0.2 volt rms sine wave (1200 Hz) to P1-4 with respect to GND. The output response shall be as specified in 3.2.1.4.

4.6.3.5 Radio input. Apply an +8 millivolt to 2 volt rms sine wave (1200 Hz) to J1-B5 with respect to J1-A4 and connect P1-3 to GND. The output response shall be as specified in 3.2.1.4 and 3.2.1.5.

4.6.3.6 Wireline input. Apply a +7.75 then a 775 millivolt rms sine wave (1200 Hz) to J1-B6 with respect to J1-A6. The output response shall be as specified in 3.2.1.4 and 3.2.1.6 for each of the specified inputs.

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4.6.4 Common connector contact continuity. This test shall be conducted with no external power connected to the interface. Measure with ohm meter the resistance between the connector contacts specified in 3.2,1.7. All measured resistances shall be as specified in 3.2.1.7.

4.6.5 Workmanship. The interfaces shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The assembly shall conform to the requirements of 3.6.

4.7 Reliability testing. The requirement for reliability test approval and the responsibility (Government or contractor) for reilability testing shall be as specified in the contract. The sample for reliability test shall consist of 3 interfaces. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.7.1 Government testing. When the Government is responsible for conducting reliability tests the contractor Prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspected and tested by-the contractor for all the requirements of the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials~ shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractors reliability tests and inspections.

5. PREPARATION FOR DELIVERY

5.1 Delivery requirements. Handling, packaging, packing, shipping, and marking of the Audio Interface Circuit Card Assembly, 9355830 shall be in accordance with Packaging Data Sheet 9355830. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The interface described herein is intended for use in the MBC. See drawing 11785850. “

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d. Applicable Packaging Data Sheet.

e. Applicable levels of preservation, packing, and marking (see 5.1).

f. Applicable stock numbers.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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